

Q1 - 25 July - Shift 1

The required height of a TV tower which can cover the population of 6.03 lakh is h . If the average population density is 100 per square km and the radius of earth is 6400 km, then the value of h will be _____ m.

*Space for your notes:***Q2 - 26 July - Shift 1**

A radio can tune to any station in 6 MHz to 10 MHz band. The value of corresponding wavelength bandwidth will be :

Space for your notes:

- (A) 4 m (B) 20 m
(C) 30 m (D) 50 m

Q3 - 26 July - Shift 2

The maximum and minimum voltage of an amplitude modulated signal are 60 V and 20 V respectively. The percentage modulation index will be :

Space for your notes:

- (A) 0.5% (B) 50% (C) 2% (D) 30%

Q4 - 27 July - Shift 1

At a particular station, the TV transmission tower has a height of 100 m. To triple its coverage range, height of the tower should be increased to

Space for your notes:

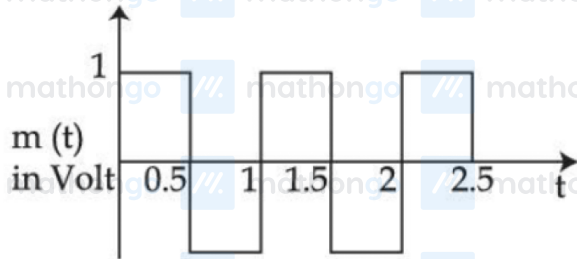
- (A) 200 m (B) 300 m
(C) 600 m (D) 900 m

Q5 - 27 July - Shift 2

Questions

MathonGo

A square wave of the modulating signal is shown in the figure. The carrier wave is given by $C(t) = 5 \sin(8\pi t)$ Volt. The modulation index is :

Space for your notes:

- (A) 0.2 (B) 0.1
(C) 0.3 (D) 0.4

Q6 - 28 July - Shift 1

In the case of amplitude modulation to avoid distortion the modulation index (μ) should be :

Space for your notes:

- (A) $\mu \leq 1$ (B) $\mu \geq 1$
(C) $\mu = 2$ (D) $\mu = 0$

Q7 - 28 July - Shift 2

A FM Broad cast transmitter, using modulating signal of frequency 20 kHz has a deviation ratio of 10. The Bandwidth required for transmission is :

Space for your notes:

- (A) 220 kHz
(B) 180 kHz
(C) 360 kHz
(D) 440 kHz

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Q8 - 29 July - Shift 1

Find the modulation index of an AM wave having 8 V variation where maximum amplitude of the AM wave is 9 V.

(A) 0.8 (B) 0.5

(C) 0.2 (D) 0.1

Space for your notes:

Q9 - 29 July - Shift 2

A modulating signal $2\sin(6.28 \times 10^6)t$ is added to the carrier signal $4\sin(12.56 \times 10^9)t$ for amplitude modulation. The combined signal is passed through a non-linear square law device. The output is then passed through a band pass filter. The bandwidth of the output signal of band pass filter will be ____ MHz.

Space for your notes:

Answer Key**Q1** (150)**Q2** (B)**Q3** (B)**Q4** (D)**Q5** (A)**Q6** (A)**Q7** (D)**Q8** (A)**Q9** (2)

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Q1 (150)

$$d = \sqrt{2Rh}$$

$$d = \sqrt{2 \times 6400 \times h \times 10^{-3}} \quad (h \text{ in m})$$

$$\text{Area} = \pi d^2$$

$$= (\pi \times 2 \times 6400 \times h \times 10^{-3}) \text{ km}^2$$

$$6.03 \times 100000 = 100 \times \pi \times 2 \times 6400 \times 10^{-3} h$$

$$h = \frac{6.03 \times 10^5}{10 \times \pi \times 128}$$

$$h = 150 \text{ m}$$

Q2 (B)

Given: Frequency $f_1 = 6\text{MHz}$

Frequency $f_2 = 10\text{MHz}$

$$\lambda_1 = \frac{c}{f_1}$$

$$\lambda_2 = \frac{c}{f_2}$$

$$\text{Wavelength bandwidth} = \lambda_2 - \lambda_1 = 20 \text{ m}$$

Q3 (B)

$$V_{\max} = 60$$

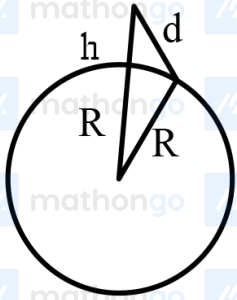
$$V_{\min} = 20$$

% modulation =

$$\left(\frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \right) 100 \Rightarrow \left(\frac{60 - 20}{60 + 20} \right) 100 \Rightarrow \left(\frac{40}{80} \right) 100$$

$$\Rightarrow 50\%$$

Q4 (D)



Let d be range

$$d^2 = (h+R)^2 - R^2$$

$$= h^2 + R^2 + 2Rh - R^2$$

$$d^2 = h^2 + 2Rh$$

as $R \gg h$ then

$$d \approx \sqrt{2Rh} \dots (1)$$

Now, if coverage is to be increased 3 times

$$3d = \sqrt{2Rh'} \dots (2)$$

Divide 2 and 1 $\frac{3d}{d} = \sqrt{\frac{2Rh'}{2Rh}}$

$$9 = \frac{h'}{h}$$

$$9h = h'$$

If $h = 100$ m then tower of height 900 m is required

Q5 (A)

$$\text{Modulation Index } \mu = \frac{A_m}{A_c} = \frac{1}{5} = 0.2$$

A_m = amp. of modulating signal

A_c = amp. of carrier wave

Q6 (A)

$$\mu = \frac{A_m}{A_c}$$

$\mu \leq 1$ to avoid distortion

because $\mu > 1$ will result in interference between
carrier frequency & message frequency.

Q7 (D)

Given

FM broadcast

Modulating frequency = 20 k Hz = f

$$\text{Deviation ratio} = \frac{\text{frequency deviation}}{\text{modulating frequency}} = \frac{\Delta f}{f}$$

$$\Rightarrow \text{frequency deviation} - \Delta f = f \times 10$$

$$= 20 \text{ kHz} \times 10 = 200 \text{ kHz}$$

$$\Rightarrow \text{Bandwidth} = 2(f + \Delta f)$$

$$= 2(20 + 200) \text{ kHz}$$

$$= 440 \text{ kHz}$$

Q8 (A)

$$\text{Modulation index: } m = \frac{A_m}{A_c}$$

$$\text{Given } 2A_m = 8$$

$$A_m + A_c = 9 \Rightarrow A_c = 5$$

$$\therefore m = \frac{4}{5} = 0.8$$

Q9 (2)

Frequencies present in output of square law device

$$2f_c, f_c + f_m, f_c, f_c - f_m, 2f_m, f_m$$

After passing through band pass filter.

$$f_c + f_m, f_c, f_c - f_m$$

$$\text{Band width} = 2f_m$$

$$\begin{aligned} &= \frac{2\omega_m}{2\pi} = \frac{6.28 \times 10^6}{3.14} \\ &= 2 \text{ MHz} \end{aligned}$$